

20.1.3 Biasing the Current Mirror

Using a resistor to set the bias current, as seen in Figs. 20.2–20.4, can result in currents that are too dependent on the power supply value and temperature. Consider the current mirror seen in Fig. 20.10a. In this design we’ve used the sizes and bias current given in Table 9.2 (the short-channel CMOS process) to select the resistor. In particular, $V_{SG} = 0.35\text{ V}$ so the gate potential is 0.65 V . Figure 20.10b shows how the reference and output currents vary if V_{DD} is swept from 900 mV to 1 V. The reference current is linearly dependent on V_{DD} (as seen in Ex. 20.1). The output current is dependent on both the reference current and the V_{DS} (λ) of M2. In general, we want a reference current that isn’t dependent on power supply or ground variations (noise). This is an important point. Consider the following example.

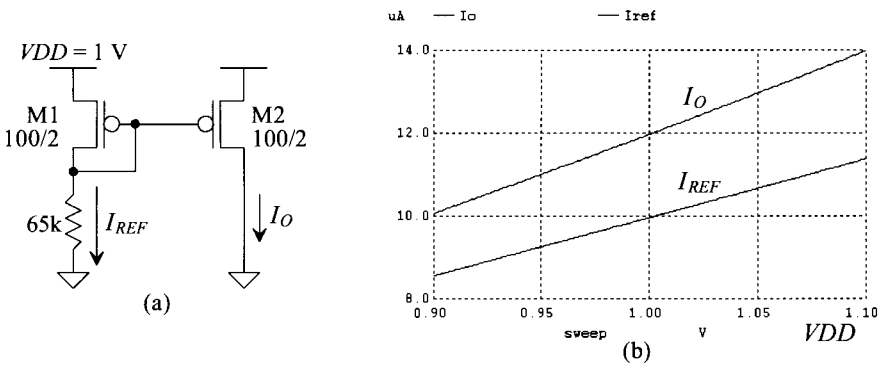


Figure 20.10 How reference and output current vary with V_{DD} .

Example 20.2

Regenerate Fig. 20.10 if the 65k resistor is replaced with an ideal $10\text{ }\mu\text{A}$ current source, Fig. 20.11a. Explain the results.

The simulation results are seen in Fig. 20.11b. Note how the output current variation with V_{DD} is much better than what is seen in Fig. 20.10. The reference current through M1 is, of course, a constant. The output current is larger than the

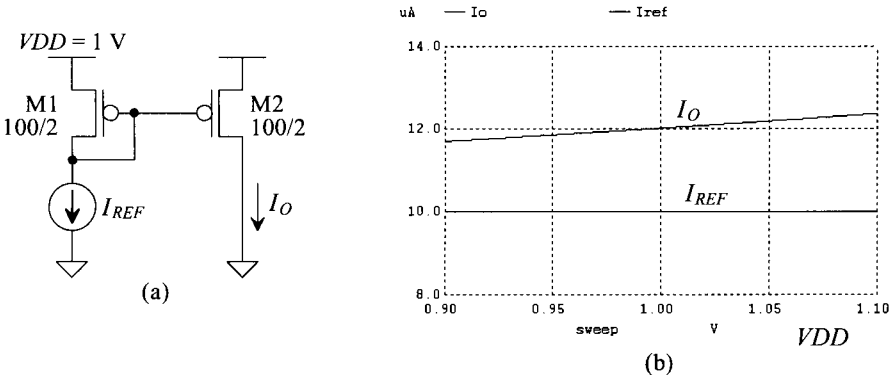


Figure 20.11 Showing that V_{DD} variations don’t affect the reference current.

reference current because $V_{SD1} < V_{SD2}$ (notice how often this point is coming up in the design of current mirrors). The reason I_o varies is due to the finite output resistance ($\lambda \neq 0$) of M2, Eq. (20.2).

The point of this example is that it is desirable to have a bias current that doesn't vary with changes in V_{DD} or ground. ■

Using a MOSFET-Only Reference Circuit

Let's try replacing the resistor bias with a MOSFET, Fig. 20.12. If we assume long-channel behavior, then

$$V_{DD} = V_{SG3} + V_{GS1} \quad (20.12)$$

or

$$V_{DD} = \sqrt{\frac{2I_{REF}}{KP_p \frac{W_3}{L_3}}} + V_{THP} + \sqrt{\frac{2I_{REF}}{KP_n \frac{W_1}{L_1}}} + V_{THN} \quad (20.13)$$

If we are going to think of M3 as a resistor, as in Fig. 20.2, so that we mirror the current in M1, then we solve for the size of M3. Using the data from Table 9.1, we get

$$5 = \sqrt{\frac{2 \cdot 20}{40 \cdot \frac{W_3}{L_3}}} + 0.9 + \sqrt{\frac{2 \cdot 20}{120 \cdot \frac{10}{2}}} + 0.8 \rightarrow \frac{W_3}{L_3} = 0.11 \approx \frac{10}{90} \quad (20.14)$$

If we are going to think of M1 as a resistor, as in Fig. 20.10a, so that we mirror the current in M3, then we solve for the size of M1 as we did in Eq. (20.14) and get

$$\frac{W_1}{L_1} \approx \frac{10}{270} \quad (20.15)$$

Figure 20.13 shows the operation of these MOSFET bias circuits with V_{DD} swept from 4.5 to 5.5 V. Note that, at 5 V, we should have a bias current of 20 μA . What we get is 14 μA . The reason that we have such a large difference is due to neglecting the output resistance of the MOSFETs (and the effects of mobility degradation). Note that the sensitivities, how the output current changes with V_{DD} , are roughly 8 $\mu\text{A}/\text{V}$. (Let's compare this to hand-calculated values of sensitivity next.)

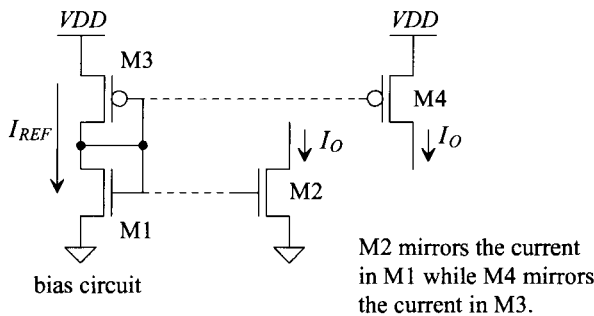


Figure 20.12 A MOSFET-only bias circuit.

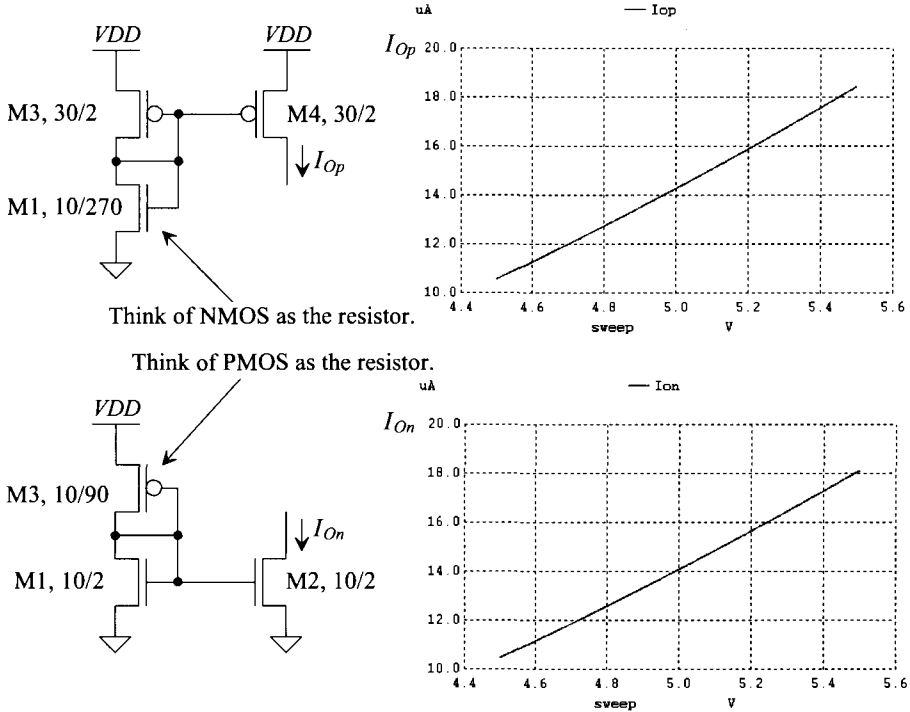


Figure 20.13 Behavior of MOSFET-only bias circuits with changes in V_{DD} .

To determine I_{REF} 's sensitivity to V_{DD} , let's take the derivative of the I_{REF} in Eq. (20.13) with respect to V_{DD} . First let's write

$$(V_{DD} - V_{THN} - V_{THP})^2 = I_{REF} \cdot \left[\sqrt{\frac{2L_3}{K P_p \cdot W_3}} + \sqrt{\frac{2L_1}{K P_n \cdot W_1}} \right]^2 \quad (20.16)$$

and next

$$\frac{\partial I_{REF}}{\partial V_{DD}} = \frac{2 \cdot V_{DD}}{K} - \frac{2 \cdot (V_{THN} + V_{THP})}{K} \quad (20.17)$$

Using the values in Table 9.1 and Eqs. (20.14) or (20.15), K is approximately $547 \times 10^3 \text{ V}^2/\text{A}$ so

$$\frac{\partial I_{REF}}{\partial V_{DD}} = 12 \text{ } \mu\text{A/V} \quad (20.18)$$

For every millivolt change in V_{DD} (around $V_{DD} = 5 \text{ V}$), we get 12 nA change in I_{REF} . As a comparison from Ex. 20.1, we can write (assuming the change in V_{GS} with V_{DD} small)

$$\frac{\partial I_{REF}}{\partial V_{DD}} \approx \frac{1}{R} = \frac{1}{200k} = 5 \text{ } \mu\text{A/V} \quad (20.19)$$

For every millivolt change in V_{DD} we get 5 nA change in bias current.

Supply Independent Biasing

Instead of putting the resistor in the drain side of the current mirror, let's consider placing it in the source side, Fig. 20.14a. One of the problems with this approach is that if we try to mirror the current in M2 using M5 we won't know the value of the mirrored current. It's not obvious how V_{GS2} and V_{GS5} are related. In (b) we add a diode connected M1 so that its current can be mirrored (by M5). The next question is how do we force the same current through both M1 and M2? Figure 20.14c shows the addition of a PMOS current mirror to do this. From (c) we can write

$$V_{GS1} = V_{GS2} + I_{REF} \cdot R \quad (20.20)$$

which can only be valid if $V_{GS1} > V_{GS2}$. To ensure that this is the case, we use a larger value of β in M2, that is, we *multiply-up* M1's β in M2 so that less gate-source voltage is needed to conduct I_{REF} . Generally, this is done by simply using a larger width in M2. The resulting circuit is called a *Beta-multiplier* reference circuit. Knowing

$$V_{GS} = \sqrt{\frac{2I_D}{\beta}} + V_{THN} \quad (20.21)$$

$$\left(\beta = KP_n \cdot \frac{W}{L} \right) \text{ and}$$

$$\beta_2 = K \cdot \beta_1 \text{ (which is satisfied by } W_2 = K \cdot W_1 \text{)} \quad (20.22)$$

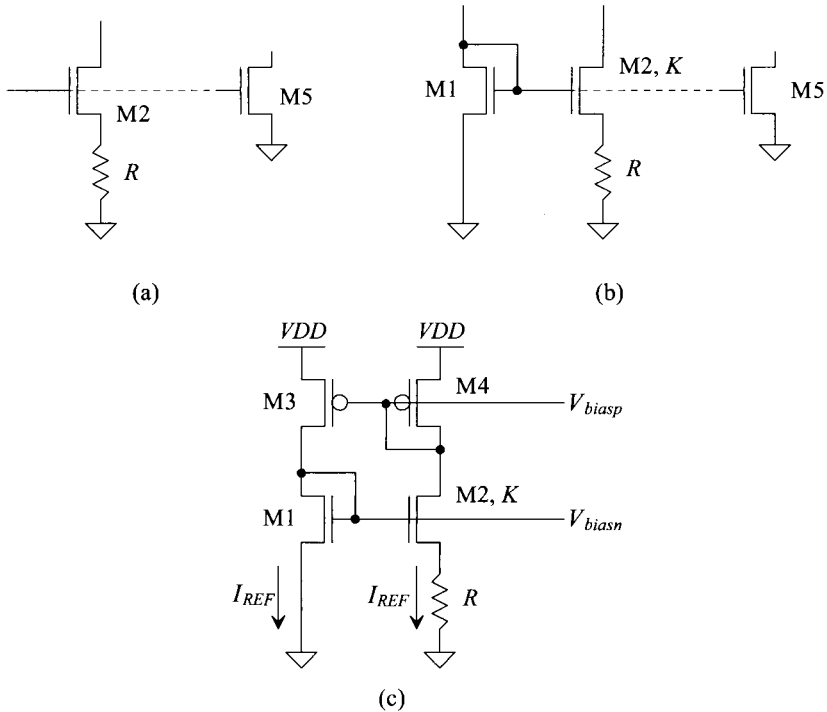


Figure 20.14 Developing the Beta-multiplier reference.

we can write

$$I_{REF} = \frac{2}{R^2 K P_n \cdot \frac{W_1}{L_1}} \left(1 - \frac{1}{\sqrt{K}}\right)^2 \quad \text{or} \quad V_{DS,sat} = V_{GS} - V_{THN} = \frac{2}{R \cdot K P_n \cdot \frac{W_1}{L_1}} \left(1 - \frac{1}{\sqrt{K}}\right) \quad (20.23)$$

It's important to note I_{REF} and V_{ovn} ($= V_{GS} - V_{THN}$), neglecting the finite output resistance of the MOSFETs which wasn't included in the derivation of Eq. (20.23), are independent of V_{DD} . Solving for R with the values given in Table 9.1 and a K of 4 gives $R = 6.5 \text{ k}\Omega$. Note that this circuit, when $K = 4$, is sometimes called a *constant- g_m* bias circuit because

$$g_m = \sqrt{2 K P_n \frac{W}{L} \cdot I_{REF}} = \frac{1}{R} \quad (20.24)$$

a constant independent of MOSFET process shifts. Figure 20.15 shows the schematic of a bias circuit based on the sizes, bias currents, and $V_{DS,sat}$ given in Table 9.1.

Note that a “start-up circuit” was included in Fig. 20.15. In any self-biased circuit there are two possible operating points: the one we just described and the unwanted one where zero current flows in the circuit. This unwanted state occurs when the gates of M1/M2 are at ground while the gates of M3/M4 are at V_{DD} . When in this state, the gate of MSU1 is at ground and so it is off. The gate of MSU2 is somewhere between V_{DD} and $V_{DD} - V_{THP}$. MSU3, which behaves like an NMOS switch, turns on and leaks current into the gates of M1/M2 from the gates of M3/M4. This causes the current to snap to the desired state and MSU3 to turn off. *Note that during normal operation the start-up circuit should not affect the Beta-multiplier's operation.* The current through MSU3 should be zero (or very small).

The Beta-multiplier is an example of a circuit that uses positive feedback. The addition of the resistor kills the closed loop gain (a positive feedback system can be stable if its closed loop gain is less than one). However, if we decrease the size of the resistor, we increase the gain of the loop and push the feedback system closer to instability. An example of when this could occur is if the parasitic capacitance on the source of M2 to ground is large (effectively shorting M2's source to ground). If the resistor, for example, is bonded out off-chip to set the current, it is likely that this bias circuit will oscillate.

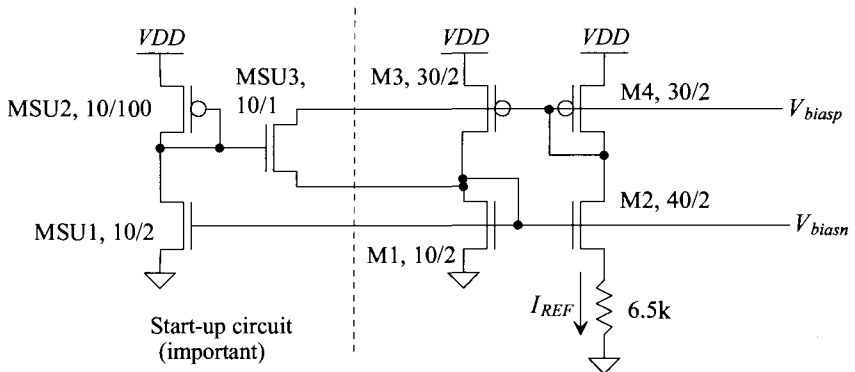


Figure 20.15 Beta-multiplier reference for biasing in the long-channel process described in Table 9.1.

Figure 20.16 shows how the reference currents through M1 and M2 vary with V_{DD} . The minimum value of V_{DD} can be estimated by looking at the minimum value of voltage across the drain-source of M3 and M1. For M3 this is $V_{SD3,sat}$ or 250 mV (because we are using the parameters from Table 9.1 in the design of this current reference). For M1 this is $V_{DS1} = V_{GS1} = 1.05$ V. We can then write

$$V_{DD\min} = V_{SD3,sat} + V_{GS1} = 1.3 \text{ V} \quad (20.25)$$

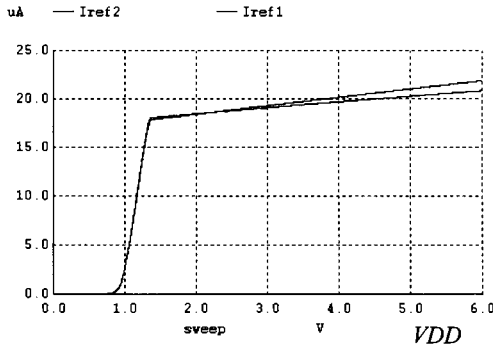


Figure 20.16 The reference currents through M1 and M2 in the Beta-multiplier.

Finally, the sensitivity of I_{REF} is directly dependent on the output resistance of the MOSFETs. From the simulations

$$\frac{\partial I_{REF}}{\partial V_{DD}} \approx 800 \text{ nA/V}$$

or almost an order of magnitude better than the previous reference circuits

Example 20.3

Estimate the voltage on the gate of M6 and its drain current in Fig. 20.17.

M3 and M4 are biased to source 20 μA of current. Since $V_{GS1} = V_{GS2} = V_{DS1}$ (and M1/M2 have the same drain current, see Fig. 20.1 and the associated discussion), it follows that $V_{GS6} = V_{DS2} = V_{GS1} = 1.05$ V. We treat M6, for biasing purposes, as if its gate were tied to the gates of M1 or M2. It follows then that $I_{D6} = 20 \mu\text{A}$. ■

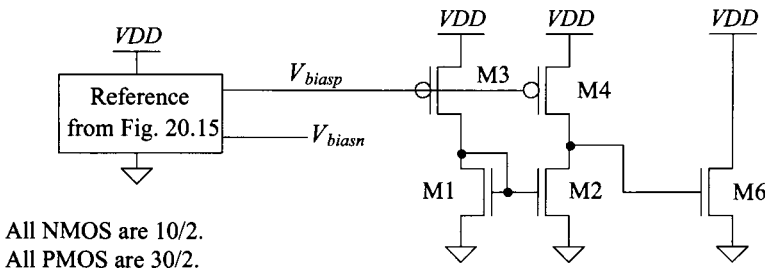


Figure 20.17 Circuit used in Ex. 20.3

20.1.4 Short-Channel Design

Figure 20.18 shows a Beta-multiplier biasing circuit based on the values given in Table 9.2 as well as simulation results showing how the reference currents vary with V_{DD} . From Eq. (20.25) and Table 9.2 we would expect $V_{DD_{min}}$ to be 400 mV. At $V_{DD} = 1$ V the reference currents are indeed 10 μ A. However, what we see is a horrible sensitivity to changes in V_{DD} . Reviewing Figs. 9.31 and 20.10 we see the low output resistance of the short-channel devices causes the drain current to change significantly with changes in drain-to-source voltage. In some analog applications this variation isn't that harmful. However, if the Beta-multiplier circuit is to behave like a true current reference, the reference currents shouldn't vary with changes in V_{DD} .

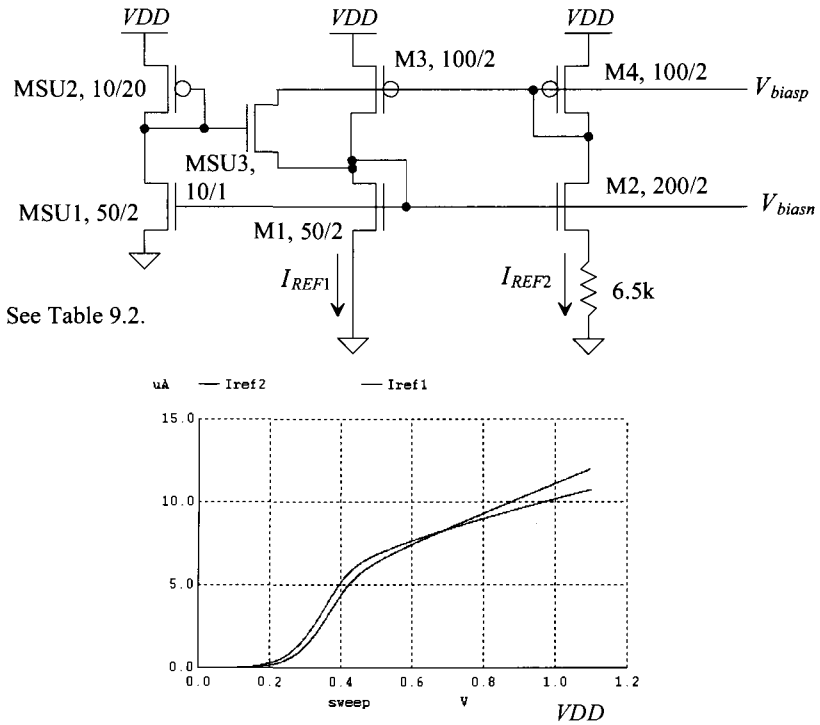


Figure 20.18 Beta-multiplier reference for short-channel design (see Table 9.2).

To reduce the sensitivity, we need to reduce the variations in the drain-to-source voltages of the NMOS devices with changes in V_{DD} . Consider adding a differential amplifier (diff-amp) to the basic Beta-multiplier seen in Fig. 20.19. Note that M4 is no longer diode-connected so its drain can move to the same potential as M2's drain, that is, V_{biasn} . The start-up circuit (required) is not shown. The idea is to use the amplifier to compare the drain voltage of M1 (V_{biasn}) with the drain voltage of M2 (V_{reg}) and regulate them to be equal. The result is an effective increase in M2's output resistance. For example, if V_{reg} is above V_{biasn} , the amplifier's output voltage increases. This drives the

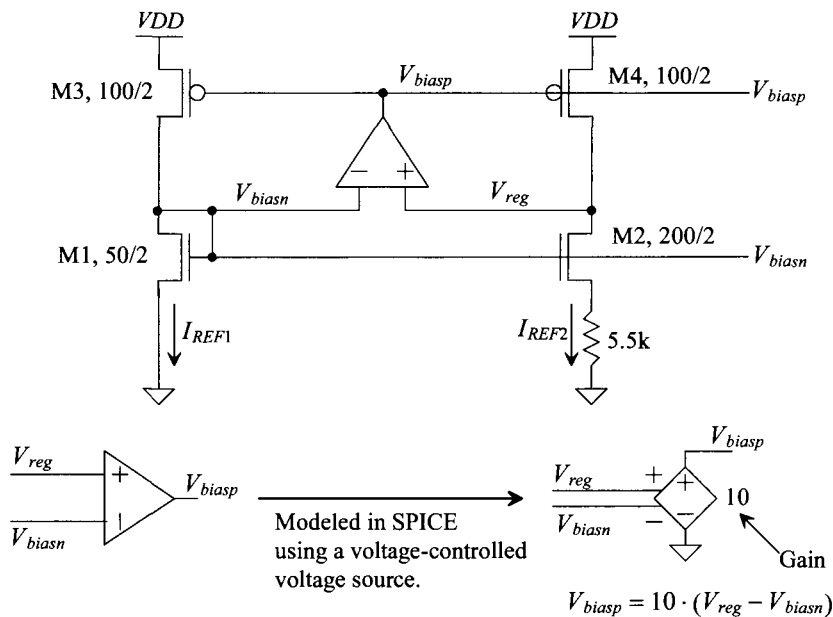


Figure 20.19 Increasing the output resistance of short-channel MOSFETs using feedback. The result, for the Beta-multiplier circuit, is better power supply sensitivity.

gate of M4 upwards, lowering the current it supplies and causing V_{reg} to drop back down. At the same time the gate of M3 is also increased, causing it to source less current. This causes a drop in V_{biasn} (the same as V_{reg} because of the symmetry as discussed in Fig. 20.1 or Ex. 20.3). Figure 20.20 shows how the reference current, in Fig. 20.19, changes with V_{DD} when the added amplifier has a gain of 10. We've lowered R to 5.5k to more accurately set the current and used, in the simulation, a voltage-controlled voltage source for the amplifier.

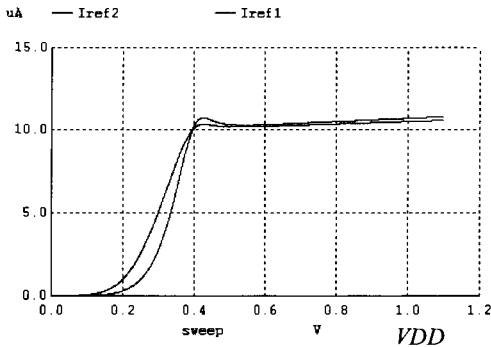


Figure 20.20 Improvement with the added amplifier.

Figure 20.21 shows a possible differential amplifier configuration. The PMOS devices form a current mirror while the NMOS devices will, when $V_{biasn} = V_{reg}$, simply mirror the current in M1. If $V_{biasn} \neq V_{reg}$ then the imbalance causes the amplifier output to swing up or down providing the desired action.

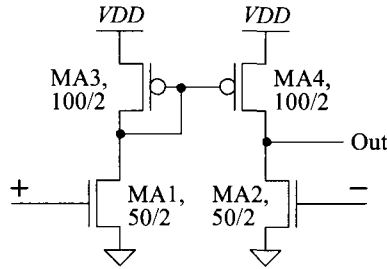
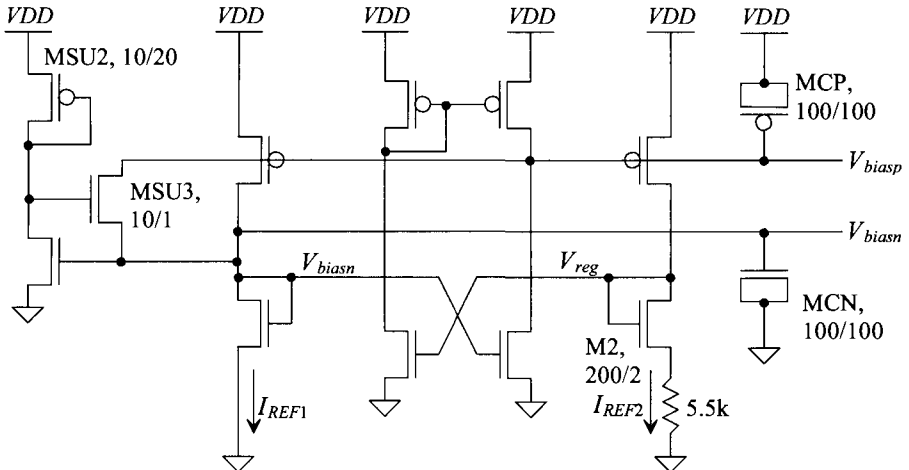


Figure 20.21 A possible implementation of the diff-amp in Fig. 20.19.

Figure 20.22 shows the improved current reference for use in a short-channel CMOS process (where devices have very low output resistances). The amplifier in Fig. 20.21 is placed in the reference, as indicated in Fig. 20.19. As with any feedback scheme, stability is of critical importance. As we'll see in the coming chapters, a feedback amplifier with only a single high-impedance node (meaning no diode-connected MOSFETs or MOSFET-source terminals are connected to the node) is straightforward to compensate (make stable). So that the reference has a single high-impedance node, that is V_{biasp} , we connected M2's gate to its drain (instead of to V_{biasn}). We get the same effect, as



All unlabeled NMOS are 50/2.
All unlabeled PMOS are 100/2.

Figure 20.22 Improved current reference for short-channel devices.
See also Sec. 23.1.3 for more information.

seen in Fig. 20.19 but the gain around the loop is reduced. To make the reference stable, we add capacitors (MCP and MCN) to the circuit. The high-impedance node is the critical point for adding the compensation capacitor (that is MCP is the critical capacitance). If the reference drives a significant number of MOSFETs (so these driven MOSFETs provide the load capacitance), then MCP and MCN may be excluded from the design. Figure 20.23 shows how the reference currents change with V_{DD} .

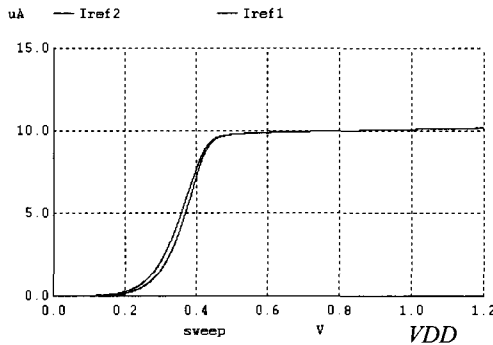
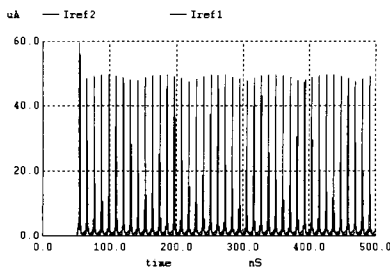


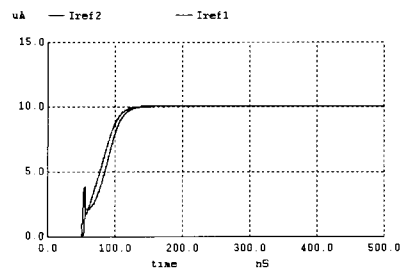
Figure 20.23 Variation of reference currents with V_{DD} for the circuit in Fig. 20.22.

An Important Note

It's extremely important to understand that together with the benefits of better power supply sensitivity are the undesired traits of feedback, that is, the potential for an unstable circuit. For example, Fig. 20.24a shows what happens to the reference currents if we remove MCP and MCN in Fig. 20.22 and then apply a step voltage to V_{DD} (that is, V_{DD} steps from 0 to 1 V at 50 ns in the simulation). Clearly, the reference is not stable and the currents oscillate. In (b) we do the same thing but with MCP and MCN present. The response shows first-order behavior, and oscillations are not present. We might think we are done with the reference and move on to the other analog circuit designs. However, further characterization of the current reference using simulations is warranted. For example, what happens if V_{DD} has a 50 mV squarewave signal coupled to it at 100 MHz? How does the reference behave with this signal (a squarewave V_{DD} that oscillates between 1 and 1.05 V at 100 MHz)? In general, no analog circuit has good power supply noise rejection at high frequencies. To remedy this, the power supply is decoupled (a large capacitor is placed from V_{DD} to ground to remove high-frequency noise).



(a) MCP and MCN not present



(b) MCP and MCN present

Figure 20.24 What happens when V_{DD} is pulsed from 0 to 1 at 50 ns.

20.1.5 Temperature Behavior

While we've been focusing on how the reference current changes with V_{DD} , it's also important to know how the current changes with temperature. Looking at the basic current mirror in Fig. 20.11a, note that if I_{REF} is a constant, independent of temperature, then the output current will also be independent of temperature. That's not to say that the MOSFET characteristics aren't changing because they are; they change at the same rate. Because they change together, the current mirror relationship is still valid and M2 mirrors the current in M1. Of course, if M1 is located at a different physical location than M2 and each is heated differently, a mismatch in the currents will result. *The point is that the temperature behavior of the reference current determines the temperature behavior of all mirrored currents.*

Using the short-channel CMOS parameters we know, from Ch. 9 (Fig. 9.30b), that the change in V_{THN} with temperature is $-0.6 \text{ mV}/^\circ\text{C}$ (for an NMOS device). For the PMOS device we can vary temperature and look at the change in V_{THP} . Using the mirror in Fig. 20.11 and looking at the V_{SG} at different temperatures results in the plots seen in Fig. 20.25. We see that it also varies at a rate of $-0.6 \text{ mV}/^\circ\text{C}$.

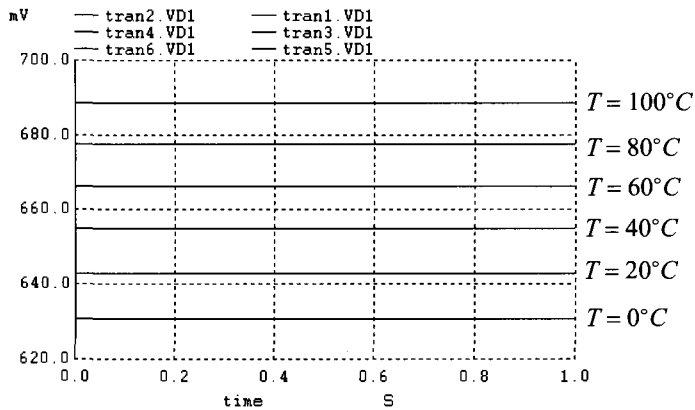


Figure 20.25 Variation in the reference gate voltage in the PMOS mirror seen in Fig. 20.11.

Resistor-MOSFET Reference Circuit

For a resistor-MOSFET reference circuit (Fig. 20.2), we can write (neglecting the MOSFET's finite output resistance)

$$I_{REF} = \frac{V_{DD} - V_{GS}}{R} \quad (20.26)$$

The change in I_{REF} with temperature is

$$\frac{\partial I_{REF}}{\partial T} = -\frac{V_{DD} - V_{GS}}{R^2} \cdot \frac{\partial R}{\partial T} - \frac{1}{R} \cdot \frac{\partial V_{GS}}{\partial T} \quad (20.27)$$

or

$$\frac{\partial I_{REF}}{\partial T} = -I_{REF} \cdot \frac{1}{R} \frac{\partial R}{\partial T} - \frac{1}{R} \cdot \frac{\partial V_{GS}}{\partial T} \quad (20.28)$$